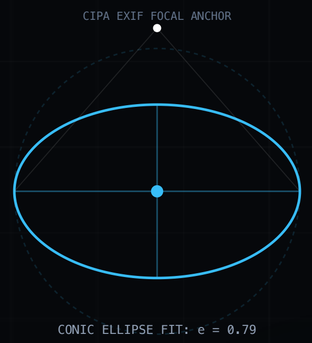
QUIET PRECISION // SYSTEM ATLAS

MessMate Dietary Intelligence

SPECIFICATION: v2.0 OPEN-WORLD
CORE DAG LATENCY: < 442 MS
VLM PROTOCOL: DETERMINISTIC (K=1, T=0.0)
INVARIANT: I1 ZERO-NUTRIENT ESTIMATION

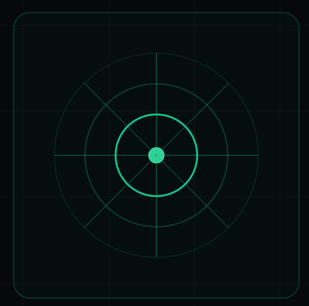
REF 01 — OPTICAL PINNING



The Metric Camera

Android hardware captures focal length and projective rim invariants, grounding 2D pixels in true 3D metric scale.

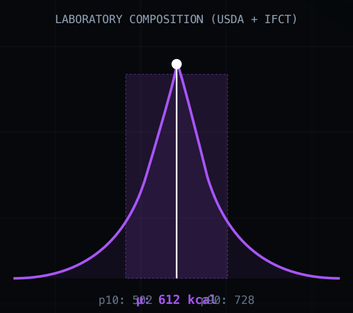
REF 02 — VISUAL RE-ID



Tableware Fingerprint

Self-supervised DINOv2 recognizes your personal bowl across weeks, separating vessel volume from visual noise.

REF 03 — CHEMICAL TRUTH



Conformal Nutrition

Zero LLM hallucinations. Calories are mathematically integrated from chemical databases with honest uncertainty bounds.

REF 04 — AMBIENT VoI



The Polite Host

Value-of-Information limits interrupts to ≤3/day. Prompts arrive post-meal, driving friction to zero by Day 14.

FRAMEWORK: **FastAPI / Python 3.11** CLIENT: **Kotlin 17 / Android SDK 36** EMBEDDING: **FastEmbed 384-D HNSW** UNIT COGS: **< $0.11 / USER / MONTH**